

Session 14:

AI That Reads - Natural Language Processing (NLP)

Today, you'll discover how AI understands and works with human language from breaking sentences into tiny pieces to figuring out meaning and even generating new text. It's like looking under the hood of a chatbot or a translation app. You'll learn about the magic (and the limits) of Natural Language Processing

We will look under the hood at **how AI processes human language**, from breaking sentences into tiny pieces to understanding meaning and generating new text. We'll use **QuillBot, Google Translate, ChatGPT/Gemini, and NotebookLM** to experience common NLP tasks.



Objectives

01

Define NLP

Define **Natural Language Processing (NLP)** and explain why human language is hard for computers.

02

The Three-Stage Pipeline

Describe the three-stage NLP pipeline:
Break → Understand → Generate

03

Breaking Down Steps

Explain key "breaking down" steps: **tokenization, stemming, lemmatization, part-of-speech tagging, named entity recognition (NER).**

04

Common NLP Tasks

Identify **common NLP tasks** (sentiment analysis, text classification, machine translation, summarisation, question answering).

05

Challenges in NLP

Recognise **challenges in NLP** (ambiguity, sarcasm, idioms, cultural context, bias).

06

Hands-On Tools

Use **QuillBot, Google Translate, ChatGPT/Gemini, and NotebookLM** to perform common NLP tasks.

07

Prompt Library

Add an NLP-focused prompt to your **Prompt Library**.

Recap: What We Learned in Session 13

Key Idea	What It Means
Domain-specific prompting	Adapt your prompt to creative, everyday, research, learning, or writing tasks.
Verification	Always check AI facts against trusted sources.
Transparency	Be honest about using AI - you are the author.
QuillBot	A tool for paraphrasing and polishing your own writing.
Responsible AI use	AI is a thinking partner, not a shortcut.

 **The most important idea from Session 13:** You can use AI for real tasks, from planning a picnic to writing a story but you stay in control, verify facts, and never submit AI work as your own.

Warm-Up: The Language Puzzle

Read these three sentences. What makes each one tricky for a computer?

Sentence 1

"I saw a man on a hill with a telescope."

Who has the telescope? The man or the viewer?

Sentence 2

"Time flies like an arrow"

Does "flies" mean the verb or the insect?

Sentence 3

"Great weather we're having!"

(said during a thunderstorm) - Is this positive or negative?

Discussion: Humans understand sarcasm, ambiguity, and wordplay naturally. But computers see only letters and spaces. How could a machine ever figure this out?

Big Question: How does AI turn messy human language into something it can "understand" and then use that to help us?



What is Natural Language Processing (NLP)?

NLP is the branch of AI that gives computers the ability to read, interpret, and generate human language.

Think of NLP as a **translator between human and machine**. You speak or write in English (or any language). NLP breaks your words into pieces the computer can analyse, then reassembles them into a useful response.

Where you see NLP every day

Spell checkers (Google Docs, Word)

Voice assistants (Siri, Alexa)

Chatbots (customer support)

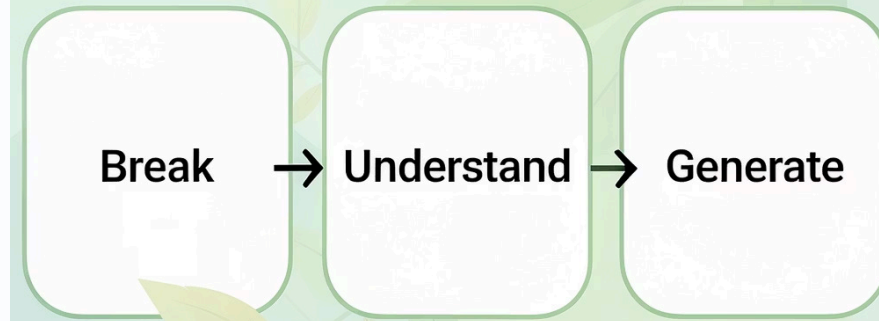
Translation apps (Google Translate)

📄 **Key insight:** NLP doesn't "understand" like you do. It finds patterns in billions of sentences and uses those patterns to predict what makes sense.

How NLP Works:

Break → Understand → Generate

Most NLP systems follow a three-stage process. This framework helps you see what's happening inside.





Break

Split text into pieces, identify word roles and entities



Understand

Figure out meaning, intent, sentiment (NLU - Natural Language Understanding)



Generate

Produce new text (NLG - Natural Language Generation)

 **Important:** These stages happen in milliseconds. The AI doesn't "think", it follows mathematical steps.

Stage 1: BREAK - Chopping Text into Edible Pieces

Before a computer can understand a sentence, it must break it down. Think of it like chopping vegetables before cooking. Here are the main "chopping" techniques:

Tokenization

Splitting text into **tokens** - words, punctuation, or sub-words.

"I love AI!" → ["I", "love", "AI", "!"]

The AI can only work with tokens, not raw strings.

Stemming

Chopping off word endings to get a **root form**.

"playing", "played", "plays" → all become "play"

Helps group similar words without needing a dictionary.

Lemmatization

Reducing a word to its **dictionary form** (more accurate than stemming).

"better" → "good" / "ran" → "run"

Preserves real meaning. Useful for search engines and chatbots.

Stage 2: UNDERSTAND (NLU - Natural Language Understanding)

Part-of-Speech (POS) Tagging

Labeling each token as noun, verb, adjective, etc.

Example: "The cat sleeps" → "The" (determiner), "cat" (noun), "sleeps" (verb)

Why it matters: Knowing a word's role helps understand sentence structure.

Named Entity Recognition (NER)

Finding and classifying **named things** - people, places, organisations, dates, quantities.

Example: "Satya Nadella, CEO of Microsoft, visited Paris on Monday"

- **Person:** Satya Nadella
- **Organisation:** Microsoft
- **Location:** Paris
- **Day:** Monday

Why it matters: Search engines, calendars, and news summarisers rely on NER.

☐ **All these "break" steps happen before the AI even tries to understand the meaning.** They turn raw text into structured data.

Once the text is broken down, the AI needs to **understand** what it means.

Intent

What does the user want?

(e.g., "What's the weather?" → intent = get forecast)

Sentiment

Positive, negative, or neutral?

Relationships

Who did what to whom?

(e.g., "John gave Mary a book." → giver = John, receiver = Mary, object = book)

Example of NLU in action: User types: "Can you turn off the living room lights?"

Intent = control device

Device = lights

Location = living room

Action = turn off

📌 **Note:** NLU is pattern matching, not true understanding. The AI doesn't "know" what a light is - it has learned that the words "turn off" + "lights" often appear together in commands.

Stage 3: GENERATE (NLG - Natural Language Generation)

After understanding, the AI may need to **generate** a response or new text.



A chatbot reply



A summary of a long
article



A translated sentence



A story or poem

Example: Input (user): "Tell me a fun fact about jellyfish."

NLU understands the intent. Then NLG generates: *"Jellyfish have no brains, yet they can detect light and move using their tentacles."*

- ❏ **Key insight:** NLG uses the same next-word prediction you learned in Session 8. It predicts the most likely sequence of tokens based on the input and its training.



Putting It All Together: A Complete Example

Sentence: "Siri, what's the weather in Tokyo tomorrow?"

Stage	What happens
BREAK	Tokenize: ["Siri", ",", "what", "'s", "the", "weather", "in", "Tokyo", "tomorrow", "?"] POS tagging: "Siri" (proper noun), "what" (pronoun), "weather" (noun), "Tokyo" (proper noun), "tomorrow" (adverb) NER: "Siri" (product), "Tokyo" (location), "tomorrow" (date)
UNDERSTAND (NLU)	Intent = get weather forecast Location = Tokyo Time = tomorrow
GENERATE (NLG)	"Tomorrow in Tokyo, it will be partly cloudy with a high of 18°C."

📄 All of this happens in less than a second!

Mini Activity 1: Match the NLP Break Step

Instructions: For each description, write which **Break** step it matches: **Tokenization**, **Stemming**, **Lemmatization**, **POS Tagging**, **NER**.

#	Description	Break Step
1	Splitting "I'm learning!" into ["I", "m", "learning", "!"]	_____
2	Reducing "running" to "run" (even if "run" is not the dictionary form)	_____
3	Reducing "better" to "good" (using a dictionary)	_____
4	Labeling "cat" as a noun and "sleeps" as a verb	_____
5	Finding "Amazon" as a company and "Jeff Bezos" as a person	_____

Mini Activity 1: Answer Key

1 - Tokenization

Breaking text into tokens (words, punctuation, contractions).

2 - Stemming

Crudely chopping endings; "running" → "run" (no dictionary check).

3 - Lemmatization

Uses a dictionary to get the proper base form.

4 - POS Tagging

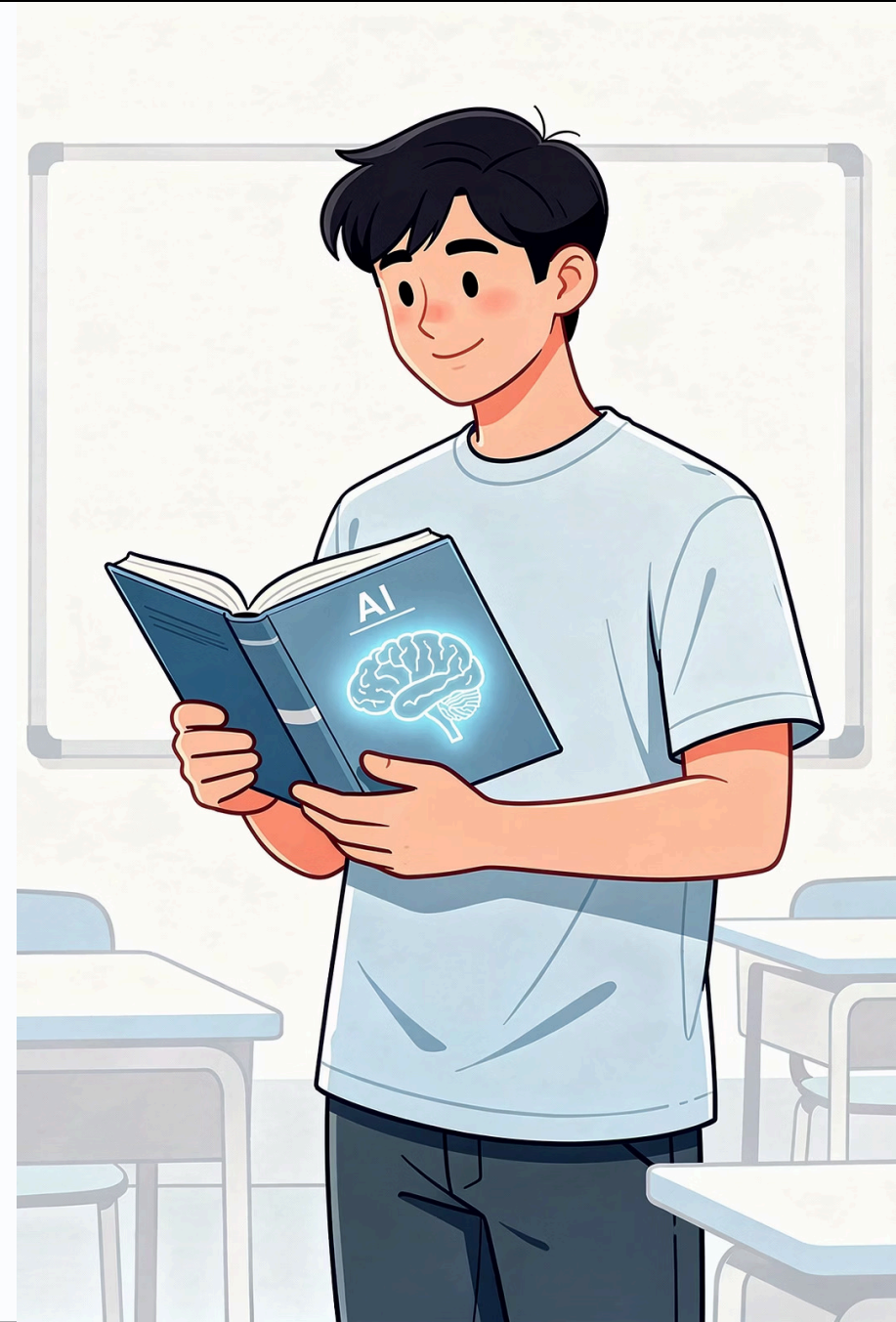
Assigns grammatical roles (noun, verb, etc.).

5 - Named Entity Recognition (NER)

Identifies and classifies named entities (people, organisations, locations, dates).

How ChatGPT Uses "Break → Understand → Generate"

Stage	What ChatGPT does
Break	Converts your sentence into tokens (small pieces of words)
Understand	Uses a transformer to analyse the context and relationships between those tokens - like a spotlight that looks at important words (you learned this as attention in Session 8)
Generate	Predicts the most likely next word in the sequence, one token at a time, to build a reply



How ChatGPT Uses "Break → Understand → Generate"

How ChatGPT works under the hood

ChatGPT uses Natural Language Processing (NLP) by leveraging a transformer-based model to convert text into tokens, analyzing context and relationships between words, and predicting the most likely next word in a sequence.

What is a "transformer" in simple words?

A transformer is a type of AI model that is very good at paying attention to the right words, even if they are far apart in a sentence. It looks at all the words at once, decides which ones matter most for understanding the meaning, and uses that to predict the next word

- ① So every time you chat with ChatGPT, you are seeing the **"Break → Understand → Generate"** Framework in action. You are no longer just using the car - you are looking inside the engine.

Common NLP Tasks

Once the Break → Understand → Generate pipeline is in place, NLP can perform many useful tasks. Here are the most common ones, with examples and the tools we'll use today.

Task	What it does	Tool We'll Use	Example
Sentiment Analysis	Detects emotion (positive/negative/neutral)	ChatGPT/Gemini	"The new cafeteria's pizza is amazing!" → Positive
Machine Translation	Converts text to another language	Google Translate	English: "Where is the library?" → Spanish: "¿Dónde está la biblioteca?"
Summarisation	Creates a shorter version	NotebookLM	A 2-page article about solar panels → 2-sentence summary



Common NLP Tasks

Task	What it does	Tool We'll Use	Example
Paraphrasing	Rewrites text while keeping meaning	QuillBot	"The quick brown fox jumps over the lazy dog." → "A fast, brown fox leaps over a sleepy dog."
Question Answering	Finds an answer to a question	ChatGPT/Gemini, NotebookLM	"Who painted the Mona Lisa?" → "Leonardo da Vinci"
Named Entity Recognition (NER)	Extracts names, places, dates	ChatGPT/Gemini	"Ada Lovelace worked in London in 1843." → Person: Ada Lovelace, Location: London, Date: 1843

 **Note:** These tasks often combine multiple stages. Today's hands-on activity will let you try each task with the recommended tool.

Challenges in NLP

Even with advanced AI, NLP faces many challenges. Understanding these helps you know when to trust NLP and when to be careful.

Challenge	What it means	Real-world example
Ambiguity	A word or sentence has multiple meanings	"The detective looked at the man with the binoculars." Who has the binoculars? The detective or the man?
Sarcasm & Irony	Words mean the opposite of what they say	"Oh great, another homework assignment." (might mean "I'm unhappy.")
Idioms	Phrases whose meaning isn't literal	"It's a piece of cake" (easy) – an NLP system might think you're talking about dessert.
Cultural Context	Language depends on where and when you live	"Football" means soccer in the UK, American football in the US.

Challenges in NLP

Challenge	What it means	Real-world example
Bias	Training data reflects stereotypes	An AI might associate "nurse" with "female" and "engineer" with "male".
Low-Resource Languages	Not enough training data	NLP works great for English, but poorly for many African or Indigenous languages.
Long-Range Dependencies	Meaning depends on words far apart	In a long story, the pronoun "he" might refer to a character introduced 10 sentences earlier.

📌 **Key takeaway:** NLP is powerful, but it's not perfect. Always verify important information, especially when sarcasm or ambiguity might be involved.

Mini Activity 2: Spot the NLP Challenge

Instructions: For each sentence, identify which **NLP challenge** is present. Choose from: **Ambiguity, Sarcasm/irony, Idiom, Cultural context, Bias.**

#	Sentence	NLP Challenge
1	"The bank was steep, so we had to climb carefully." (bank = river side, not money)	_____
2	"Break a leg!" (said to a performer before a show)	_____
3	"That's just what I needed – another pop quiz." (said after a pop quiz is announced)	_____
4	A chatbot trained on American English tells a British user to "put your sweater in the trunk" (instead of "boot").	_____
5	An image captioning AI always labels people with darker skin as "cook" and lighter skin as "chef".	_____

Mini Activity 2: Answer Key

1 - Ambiguity

The word "bank" has two meanings (river side vs. financial institution).

2 - Idiom

"Break a leg" is a theatrical idiom meaning "good luck" – literal meaning would be nonsense.

3 - Sarcasm/irony

The speaker actually means the opposite: they are unhappy about another quiz.

4 - Cultural context

"Trunk" (car storage) in US English is "boot" in UK English.

5 - Bias

The AI learned a stereotype from its training data.

NLP in the Real World



Grammar Checkers

e.g., Microsoft Editor

*POS tagging, parsing,
NLG*

Suggests changing
"She go to school" →
"She goes to school"



Customer Support Chatbots

*Intent recognition, NER,
NLG*

"I want to return a red sweater" → chatbot
identifies intent = return,
product = sweater, color
= red



Smart Reply in Email

e.g., Gmail

NLU + NLG

Offers one-tap replies:
"Yes, I'll be there" or
"No, thank you"



Voice Search on Smart Speakers

Speech-to-text + NLU

"Alexa, add 'milk and eggs' to my shopping list"

NLP in the Real World



Plagiarism Detection

e.g., Turnitin — *Text matching, tokenization*

Compares student essays against billions of web pages



Medical Record Analysis

NER, classification

Extracts "**patient name**", "**diagnosis**", "**medication**" from doctor's notes



Social Media Moderation

Sentiment analysis, classification

Flags a comment containing hate speech or bullying



Personalised News Feed

Text classification, summarisation

Groups articles by topic (sports, politics, tech) and shows you a summary



Hands-On Activity: Perform Common NLP Tasks

Goal: Perform five common NLP tasks using the tools we've focused on: **QuillBot, Google Translate, ChatGPT/Gemini, and NotebookLM.** You'll experience sentiment analysis, translation, summarisation, paraphrasing, and named entity recognition.

Hands-On Activity: Perform Common NLP Tasks



QuillBot

quillbot.com



Google Translate

translate.google.com



ChatGPT / Gemini

chat.openai.com or
gemini.google.com



NotebookLM

notebooklm.google.com

Sample text (use for multiple tasks): *"Starfish, also known as sea stars, are marine invertebrates found in oceans all over the world. They have five or more arms and can regenerate lost limbs. Starfish use tiny tube feet to move and to open shellfish like clams and mussels. They do not have a brain or blood; instead, they use seawater to pump nutrients through their bodies."*

Task 1: Sentiment Analysis (ChatGPT/Gemini)

Goal: Determine if a piece of text is positive, negative, or neutral.

- 1 In ChatGPT or Gemini, type:
"Perform sentiment analysis on the following text. Tell me if it is positive, negative, or neutral, and explain why in one sentence."
- 2 Paste this sentence: *"I tried the new blueberry muffin from the school cafeteria, and it was absolutely delicious – best one yet!"*
- 3 Write down the AI's answer.
- 4 Now try a negative sentence: *"The math homework took me two hours and I still don't understand half of it."*
- 5 Compare the two responses.

 **Reflection:** Did the AI correctly identify the sentiment? Was it ever unsure?

Task 2: Machine Translation (Google Translate)

Goal: Translate text from English to another language and back.

- 1 Open Google Translate.
- 2 Set source language to "English" and target language to "Spanish".
- 3 Type or paste: *"Where can I find a good book about space exploration?"*
- 4 Write down the Spanish translation.
- 5 Now copy the Spanish translation and paste it back into Google Translate, changing the source to "Spanish" and target to "English".
- 6 Compare the back-translated English with the original sentence.

 **Reflection:** Did the meaning change? Try the same with an idiom: *"It's raining cats and dogs."* What happens?

Task 3: Summarisation (NotebookLM)

Goal: Create a short summary of a longer text using a source-grounded tool.

- 1 In NotebookLM, create a new notebook called "NLP Tasks".
- 2 Add the **sample text about starfish** as a source (copy-paste as text source).
- 3 In the chat box, ask:
"Summarise this text in two sentences for a 5th grader."
- 4 Write down the summary.
- 5 Now ask a more specific question: *"What are tube feet and what do starfish use them for? Answer in one sentence."*

 **Reflection:** How is NotebookLM's summary different from what ChatGPT might produce without the source?

Task 4: Paraphrasing (QuillBot)

Goal: Rewrite a sentence while keeping the same meaning.

- 1 Open QuillBot.
- 2 Copy the first sentence of the sample text: *"Starfish, also known as sea stars, are marine invertebrates found in oceans all over the world."*
- 3 Paste it into QuillBot and click "Paraphrase".
- 4 Try at least two different modes: "Standard", "Fluency", "Creative".
- 5 Write down your favourite paraphrased version.
- 6 Compare it with the original. Did the meaning change?

 **Reflection:** When would paraphrasing be useful? (e.g., avoiding repetition, simplifying language, improving clarity.)

Task 5: Named Entity Recognition (ChatGPT/Gemini)

Goal: Extract people, places, organisations, and dates from a sentence.

- 1 In ChatGPT or Gemini, type: *"Perform named entity recognition (NER) on the following sentence. List each entity and its type (person, location, organisation, date, etc.)."*
- 2 Paste this sentence: *"Marie Curie was born in Warsaw, Poland in 1867. She later moved to Paris, France, where she discovered radium and polonium."*
- 3 Write down the entities the AI finds.
- 4 Try your own sentence: *"My family is planning a trip to Tokyo, Japan next summer to visit the Ghibli Museum."*

 **Reflection:** Did the AI correctly identify all entities? What about "Ghibli Museum" - is that an organisation or a location?

Step 6: Add to Your Prompt Library

Choose **one** of the prompts you used today (from sentiment analysis, NER, summarisation, etc.) and add it to your Prompt Library. Use the flexible format:

Field	Your Entry
Date	Today's date
Goal	(e.g., "Perform named entity recognition on a historical sentence")
AI Tools Used	(e.g., ChatGPT)
Final Prompt	(copy the exact prompt you used)
Elements Used	(e.g., role-based, zero-shot)
What Worked	(your observation)

Summary:

Key Term	What It Means
Natural Language Processing (NLP)	AI that works with human language – reading, understanding, generating.
Break → Understand → Generate	The three-stage pipeline of NLP (tokenization, stemming, lemmatization, POS tagging, NER → NLU → NLG).
Tokenization	Splitting text into small pieces (tokens).
Stemming	Crudely chopping word endings to get a root.
Lemmatization	Reducing a word to its dictionary form.

Summary:

Key Term	What It Means
POS Tagging	Labelling words as noun, verb, adjective, etc.
Named Entity Recognition (NER)	Finding and classifying names, places, dates, etc.
NLU (Natural Language Understanding)	Figuring out intent, sentiment, and meaning.
NLG (Natural Language Generation)	Producing new text.
Challenges in NLP	Ambiguity, sarcasm, idioms, cultural context, bias.

- 📌 **The most important idea from Session 14:** NLP is a pipeline that **breaks** language into pieces, **understands** patterns, and **generates** responses but it's all pattern matching, not true understanding. That's why you must always verify, especially for sarcasm, idioms, and important facts.



Homework

Part 1: Quick Matching

Match each term to its correct description.

#	Term	Letter	Description
1	Tokenization	A	Labelling words as noun, verb, etc.
2	NER	B	Splitting text into small pieces
3	POS Tagging	C	Finding people, places, and dates
4	Lemmatization	D	Reducing "better" to "good"
5	Stemming	E	Reducing "running" to "run" (crude)

Homework

Part 2: Identify the NLP Stage

For each scenario, write whether it belongs to **Break**, **Understand (NLU)**, or **Generate (NLG)**.

Scenario	Stage
A chatbot splits "What's the weather?" into ["What", "'s", "the", "weather", "?"]	_____
A voice assistant figures out that "Set a timer for 5 minutes" means intent = set_timer, duration = 5	_____
A translation app writes "¿Dónde está la biblioteca?" after reading "Where is the library?"	_____

Homework

Part 3: NLP Challenge Spotting

Read the sentence and write which NLP challenge it shows: **Ambiguity, Sarcasm, Idiom, Cultural context, Bias.**

"The proposal was dead on arrival." (meaning: it had no chance of being accepted)

Part 4: Add to Your Prompt Library

Choose **one** of the tasks you completed in the hands-on activity (sentiment analysis, translation, summarisation, paraphrasing, or NER). Write a prompt that you could use to perform that task again in the future. Run it in the appropriate tool (ChatGPT/Gemini, Google Translate, QuillBot, or NotebookLM). Write down the AI's response.

Add the prompt and the response to your Prompt Library (Date, Goal, Final Prompt, What Worked).

Next Session Preview

AI That Sees - Computer Vision

In next session, we will explore how AI "sees" the world - recognising objects, faces, and even emotions in images and videos. You'll also learn about computer vision in self-driving cars, medical scans, social media filters, etc.

